

COLLEGE OF ENGINEERING



ERIC G. MEYER, PH.D.

Associate Professor,
Biomedical Engineering

Research Program Officer,
Healthcare Violence Reduction Center
(HVRC)

Director, Wearable Technology Innovation
Center (WTIC)

Director, Experimental Biomechanics
Laboratory (EBL)

RESEARCH EXPERTISE

Biomedical Engineering

Biomechanical Analysis

Impact Biomechanics

Orthopedic Implants

Motion Capture

ABOUT

Dr. Meyer directs the EBL collaborations with clinicians and the medical device industry to develop preventative and regenerative treatments for bone and soft tissue damage and disease. The multidisciplinary WTIC is a partnership between LTU faculty and industry, academic, non-profit and government partners to bring research on sports injuries, workplace safety, functional aging and productivity into engineering design and innovation. The HVRC provides use-inspired research and innovative solutions around the issue of workplace violence prevention.

RESEARCH

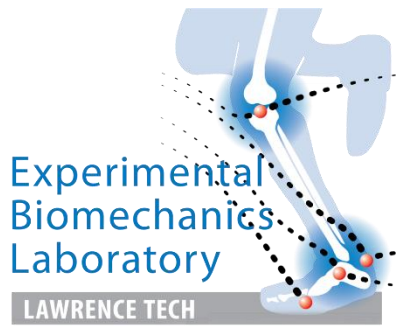
- 2023 - ONGOING **WORKPLACE VIOLENCE PREDICTIVE ANALYTICS IN HEALTHCARE**
[Funding: Michigan Hospital Association | Community Foundation of Southeast MI](#)
This project is to assess the impact of violence in healthcare settings by analyzing various sources of incident data and relevant predictor variables. The predictive analytics are applied for patient screening, emergency response wearables, AI enhanced security, robotics, worker training technology and clinical facility design.
- 2016 - ONGOING **WEARABLE TECHNOLOGY INNOVATION CENTER (WTIC) TO PROMOTE OPEN COLLABORATION FOR HUMAN ENHANCEMENT**
[PI of unfunded proposal for National Science Foundation Regional Engines: Type 1](#)
Publication: Peponis J, Meyer EG. The Fifth Element of Biomedical Engineering is Innovation: A Quintessential Design Studio Course Focusing on the "Wearable Technology" Ecosystem. ASSE Annual Conf Proc, 2025.
- 2015 - ONGOING **OSSEODENSIFICATION FOR ENHANCEMENT OF BIOMECHANICAL STABILITY AND BONE-IMPLANT CONTACT**
[Funding from Versah[®] and founder Dr. Salah Huwais](#)
Publication: Huwais S, Meyer EG. A Novel Osseous Densification Approach in Implant Osteotomy Preparation to Increase Biomechanical Primary Stability, Bone Mineral Density and Bone to Implant Contact. International J Oral & Maxillofacial Implants 2017;32:27-36. *304 Citations
- 2014 - ONGOING **BIOTECHNOLOGY FOR QUANTIFIED SELF**
[Funding from Kern Family Foundation Entrepreneurial Engineering Network](#)
Publication: Meyer EG, Nasir M. Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the "Quantified Self" social movement. ASEE Annual Conf Proc. 2015.

EDUCATION

- 2008- 2010 **POST-DOCTORAL RESEARCH FELLOW, SOFT TISSUE MECHANOBIOLOGY RESEARCH GROUP**
[Trinity College Centre for Bioengineering, Dublin, Ireland](#)
Meyer EG, Buckley CT, Thorpe SD, Kelly DJ. Low oxygen tension is a more potent promoter of chondrogenic differentiation than dynamic compression. J Biomech. 2010; 43: 2516-2523. *121 Citations
- 2004 - 2008 **PH.D., MECHANICAL ENGINEERING**
[Michigan State University](#)
Meyer EG, Haut RC. Biomechanical response of the knee in sports injury scenarios. Chapter in: Knee Joints: Kinematics, Injury Types and Treatment Options. Editor: Mascarenhas R. Nova, 2012 ISBN: 978-1-61942-268-1.
- 2002 - 2004 **B.S. & M.S., ENGINEERING MECHANICS (BIOMECHANICS)**
[Michigan State University](#)
Sports Biomechanics Consultant - Evaluate biomechanical characteristics of athletic shoes for Runner's World Magazine Shoe Guide or directly for shoe companies.

HONORS

- 2023 **LTU RESEARCH DAY**
[Honoree, Presidential Colloquium](#)
Presentation on "Disruptive Biotechnology: Past and Future Promises of Biomedical Engineering and Wearables." DOI:10.13140/RG.2.2.21335.24484
- 2017 **OSSEODENSIFICATION WORLD SYMPOSIUM**
[Invited Speaker](#)
Session on the "Biomechanical Evidence for a novel implant osteotomy preparation technique that increases implant stability."
- 2011 **IOC WORLD CONFERENCE ON PREVENTION OF INJURY & ILLNESS IN SPORT**
[Invited Speaker](#)
Session on "Solving the mystery of non-contact ACL injuries - how do they happen?"



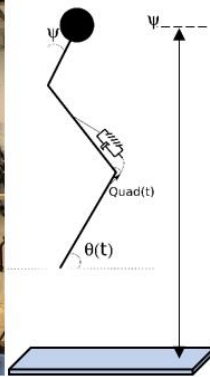
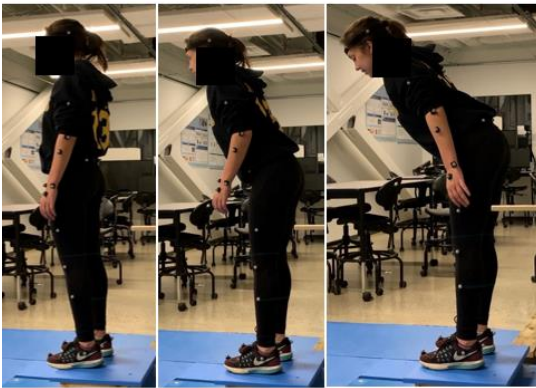
The [Experimental Biomechanics Laboratory \(EBL\)](#) at LTU is located in J134, Taubman Complex with open/flexible space for teaching and research, including a state of the art motion tracking system and other biomechanics equipment for studying many human movement activities.

The goal is to provide training to engineering and premedical students and to advance the boundary of knowledge through translational research and wearable

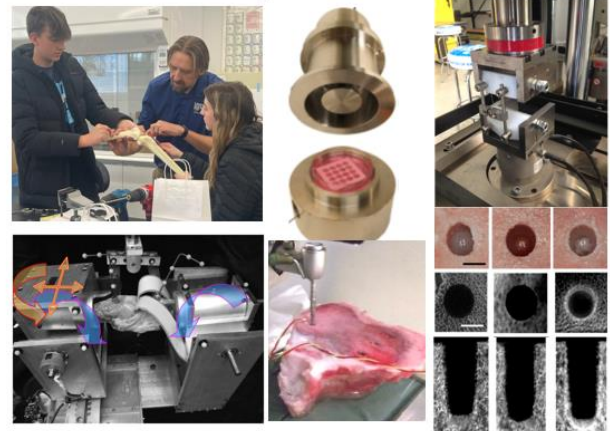
technology design with *in vivo* testing of human and animal subjects as well as cadaver and surrogate *in vitro* models and computer simulations to understand normal health, acute injury and degenerative mechanisms.



Injury Mechanisms



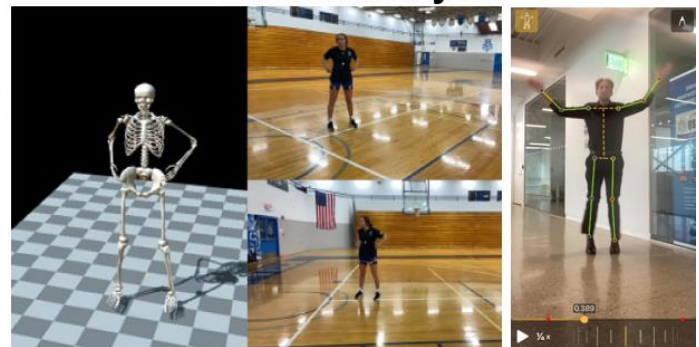
Orthopedics



Wearable Technology



Motion Analysis



Dr. Meyer received his PhD in Mechanical Engineering (Biomechanics) from Michigan State University. The primary topic of his research was lower extremity biomechanics during sports injury. He completed his post-doc at the Centre for Bioengineering at Trinity College Dublin, Ireland in cartilage tissue engineering and mechanobiology. He was an invited speaker in the session "Solving the mystery of non-contact ACL injuries - how do they happen?" at the 2011 *IOC World Conference on Prevention of Injury & Illness in Sport*, in Monte-Carlo, Monaco. Meyer has published a book chapter and 44 peer-reviewed journal and conference proceeding articles that have received more than 2600 peer citations. He has given more than 100 conference, workshop and invited presentations, including as the [Presidential Colloquium](#) speaker for the [2023 LTU Research Day](#).

"My research on sports injury and biomechanical risk overlaps with engineering design projects that allow youth and recreational athletes to track their movement to customize training using wearable sensors. As an engineering professor at Lawrence Tech, I embrace our motto:

Theory and Practice

This approach has been the culmination of my lifelong dream to protect athletes from injuries. There are 250,000 young, healthy athletes that suffer an ACL tears in the US every year. On the other hand, many of these injuries are due to uncontrolled movements that can be predicted and prevented through proper training. Just as devices like "Fitbit" have encouraged millions of people to be more active, I hope technology can bring biomechanics to underserved populations to improve their lives."